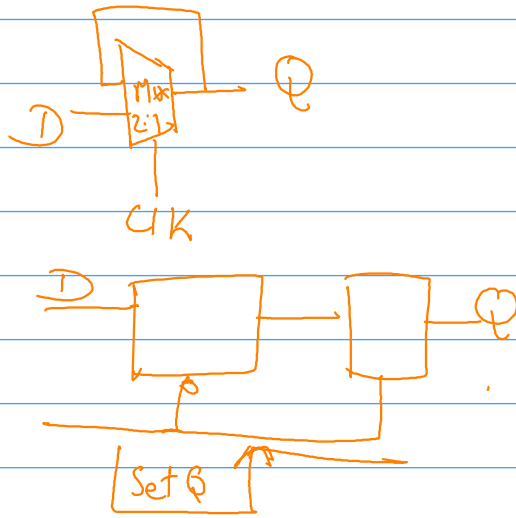


Assignment 2

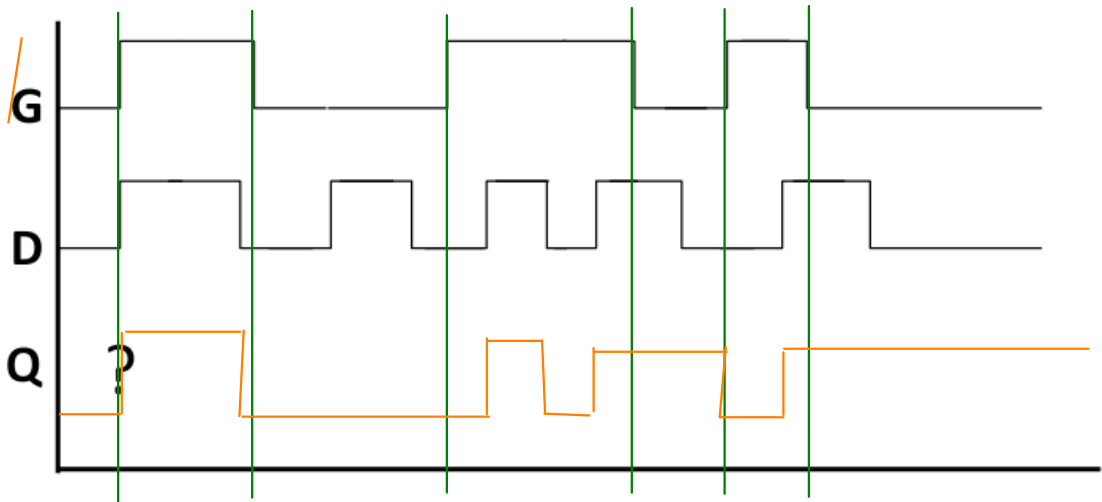
Q-1 D FF using 2:1 MUX.

⇒

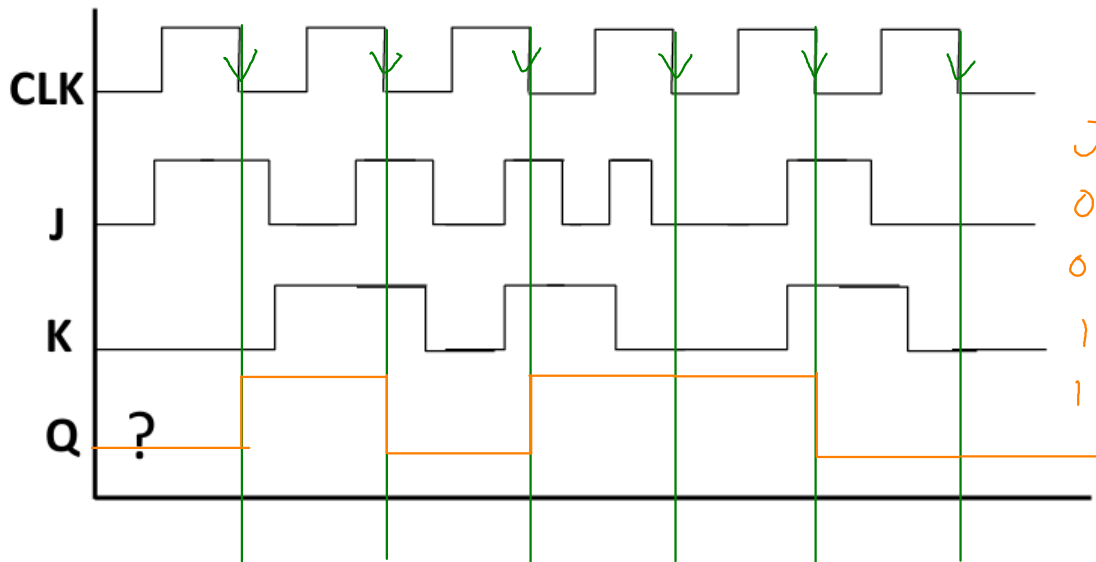


Q-2

2) Complete the below timing diagram for gated D Latch



3) Complete the below timing diagram for falling edge triggered JK FF.



J	K	Output
0	0	Init Q
0	1	Reset 0
1	0	Set 1
1	1	Toggle Q'

Q-4 Latch vs Flip flop

Controlled by enable

Edge triggered - Control by clk

Q-5 All flip flop conversion

Q	Q _{next}	SR	JK	D	T	
0	0	00	0,x	0	0	D: Q _{next} = D
0	1	10	1,x	1	1	T: Q _{next} = T ⊕ Q
1	0	01	x,1	0	1	JK: Q _{next} = JQ' + K'Q
1	1	11	x,0	1	0	SR: Q _{next} = S + RQ

Q-6 Convert JK Flip Flop to XY FF

X	Y	Q
0	0	Q
0	1	Q
1	0	Q'
1	1	Q'

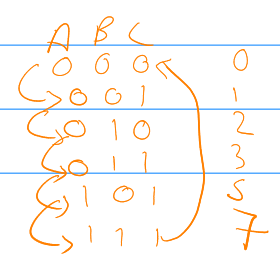
=>

TT of JK

X	Y	J	K	Q
0	1	0	0	Q
0	0	0	1	Q
1	1	1	0	Q'
1	0	1	1	Q'

J = X
K = Y'

Q-7 Design a counter which counts 0,1,2,3,5,7,0... using above XY FF and NOR gate



Max bit 3

Q	Q ⁺	X	Y
0	0	0	0
0	1	1	1
1	0	0	0
1	1	0	1

A B C	ABC ^{*(Next)}	X _A	Y _A	X _B	Y _B	X _C	Y _C
0 0 0	0 0 1	0	0	0	0	1	1
0 0 1	0 1 0	0	0	1	1	0	0
0 1 0	0 1 1	0	0	0	1	1	1
0 1 1	1 0 0	1	1	0	0	0	0
1 0 0	0 0 0	0	0	0	0	0	0
1 0 1	1 1 1	0	1	1	1	0	1
1 1 0	0 0 0	0	0	0	0	0	0
1 1 1	0 0 0	0	0	0	0	0	0

Unused
reset
counter

Solve X_A

A \ BC	00	01	11	10
0	0	0	1	0
1	0	0	0	0

$X_A = BC$

Solve Y_A

A \ BC	00	01	11	10
0	0	0	1	0
1	0	1	0	0

$Y_A = \bar{A} + \bar{B}C \approx \overline{A+BC}$

Solve X_B

A \ BC	00	01	11	10
0	0	1	0	0
1	0	1	0	0

$X_B = C$

Solve Y_B

A \ BC	00	01	11	10
0	0	1	0	1
1	0	1	0	1

$Y_B = \bar{B} + \bar{C}$

Solve X_C

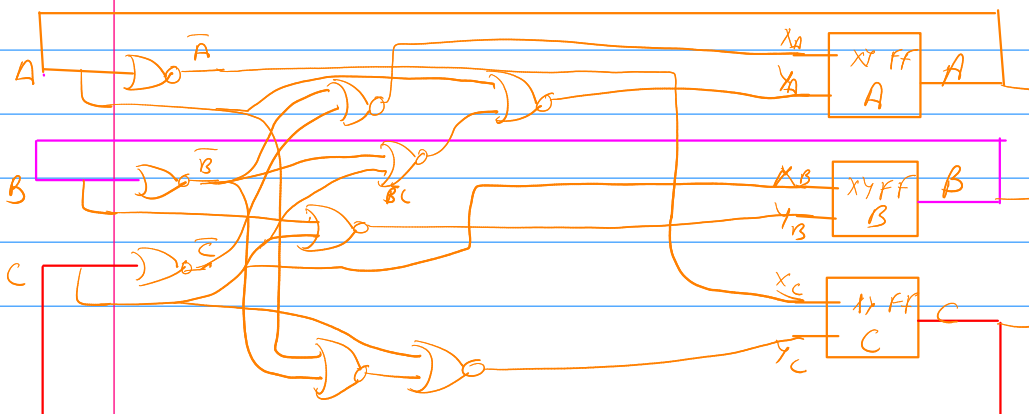
A \ BC	00	01	11	10
0	0	0	0	0
1	0	0	0	0

$X_C = \bar{A}$

Solve Y_C

A \ BC	00	01	11	10
0	1	0	0	1
1	0	1	0	0

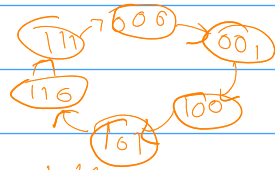
$Y_C = \bar{C} + A\bar{B} \approx C + \bar{A}B$



Q-8 Counter 0, 1, 4, 5, 6, 7, 0 and repeat

⇒

fsm



Q	Q(t+1)	T
0	0	0
1	0	1
0	1	0

Excitation table

A	B	C	A ⁺	B ⁺	C ⁺	T _A	T _B	T _C
0	0	0	0	0	1	0	0	1
0	0	1	1	0	0	1	0	1
0	1	0	0	0	0	0	1	0
0	1	1	0	0	0	0	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

BC	00	01	11	10
A	0	0	0	0
1	0	0	0	0

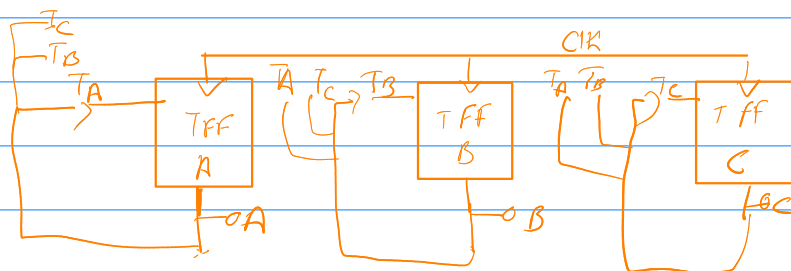
BC	00	01	11	10
A	0	0	1	1
1	0	1	1	0

BC	00	01	11	10
A	0	1	1	0
1	1	1	1	1

$$T_A = A'B'C + ABC$$

$$T_B = A'B + AC$$

$$T_C = B' + C + AB$$



Q-9

Decode Counter using T FF

* Next

4 bit 0 → 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8 → 9 → 10 → 11 → 12 → 13 → 14 → 15 → 0

⇒

A	B	C	D	M _n	A*	B*	C*	D*	T _A	T _B	T _C	T _D
0	0	0	0	1	0	0	0	1	0	0	0	1
0	0	0	1	2	0	0	1	0	0	0	1	1
0	0	1	0	3	0	0	1	1	0	0	0	1
0	0	1	1	4	0	1	0	0	0	1	1	1
0	1	0	0	5	0	1	0	1	0	0	0	1
0	1	0	1	6	0	1	1	0	0	0	1	1
0	1	1	0	7	1	0	0	0	1	1	1	1
1	0	0	0	8	1	0	0	1	0	0	0	1
1	0	0	1	9	1	0	1	0	0	0	1	1
1	0	1	0	10	0	0	0	0	1	0	1	0
1	0	1	1	11	0	0	0	1	1	0	1	1
1	1	0	0	12	0	0	0	0	1	1	0	0
1	1	0	1	13	0	0	0	0	1	1	0	1
1	1	1	0	14	0	0	0	0	1	1	1	0
1	1	1	1	15	0	0	0	0	1	1	1	1

Solve T_A

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	0	1	1
10	1	1	1	1

$$T_A = BCD + AC + AB'$$

Solve T_B

AB \ CD	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	1	1	1	1
10	0	0	0	0

$$T_B = A'CD + AB$$

Solve T_C

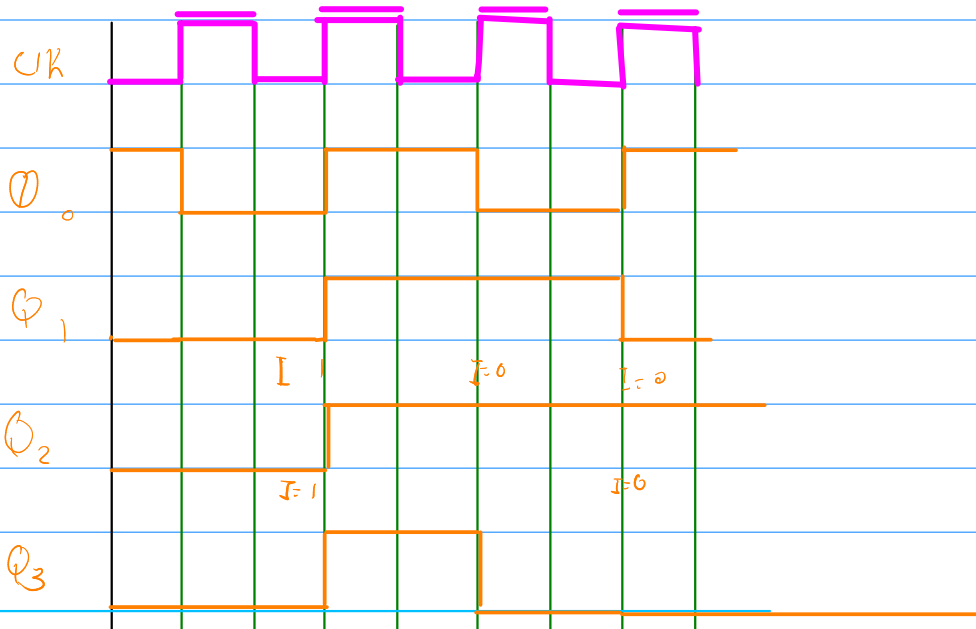
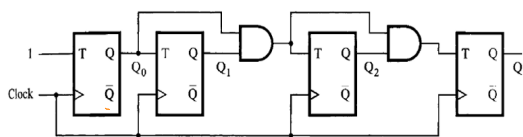
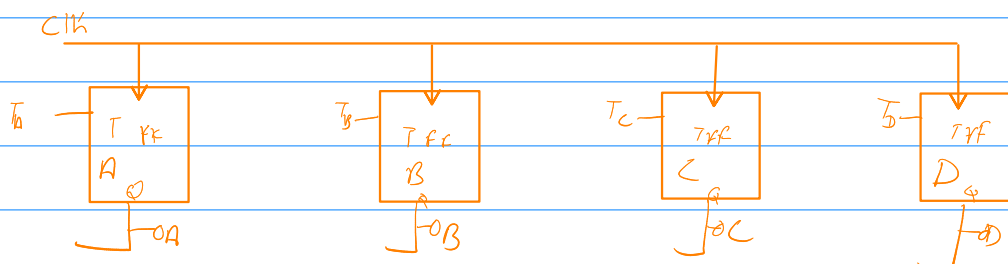
AB \ CD	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	0	1	1
10	0	1	1	1

$$\begin{aligned} T_C &= AB'D + A'D + AC \\ &= D(A' + AB') + AC \\ &= DA' + DB + AC \end{aligned}$$

Solve T_D

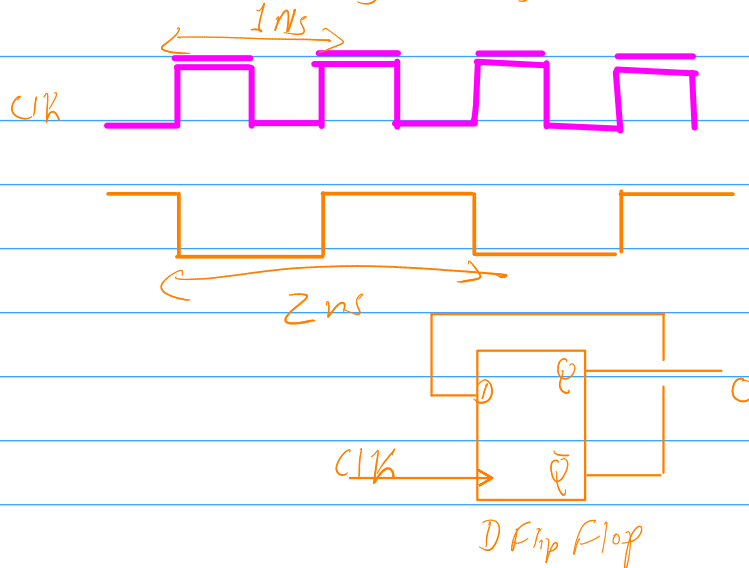
AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	1	1	0
10	1	1	1	0

$$\begin{aligned} T_D &= A' + D + A'BC' \\ &= A'(1 + BC') + D \\ T_D &= A' + D \end{aligned}$$



Q-11 frequency $\div 2$ using D FF

$\Rightarrow$ Clock takes eg 1ns, then it will take 2ns



$$f = \frac{1}{t}$$

$$\frac{f_{\text{in}}}{2} = \frac{1}{t}$$

$$f_{\text{out}} = \frac{1}{t} \times 2$$

12) Calculate the o/p frequency for below:

